Multi-Scale Regime Detection with Fuzzy Aggregation for Adaptive Sector Portfolio Management

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Abstract

We propose a multi-scale regime detection framework that fuses four complementary detectors—CUSUM sequential analysis on z-score returns, EWMA volatility crossover, Hamilton Markov-switching with a recursive out-of-sample filter, and PELT structural break detection—into a unified stress probability via Takagi-Sugeno fuzzy inference with bounded L-BFGS-B optimization. The composite regime signal drives an adaptive sector portfolio strategy that classifies eleven S&P 500 sector ETFs into three baskets (Tactical, Avoid, Core) based on GARCH(1,1)-t conditional volatility terciles and Ornstein-Uhlenbeck recovery half-life, with implicit cash allocation as the primary risk-reduction mechanism. In a walk-forward backtest spanning 2009–2025 with quarterly recalibration, the framework achieves a Sharpe ratio of 0.78, a maximum drawdown of -19.60% (versus -41.71% for SPY), and a Calmar ratio of 0.59, while capturing 98% of the benchmark return at 73% of its volatility. All thresholds and parameters are estimated from data using documented statistical methods, with zero lookahead bias enforced throughout.

Keywords: regime detection, fuzzy inference, portfolio management, CUSUM, Markov switching, GARCH, walk-forward validation, implicit cash allocation

1 Introduction

Financial markets exhibit regime-dependent behavior characterized by distinct periods of calm, transition, and stress (**?**). Traditional portfolio strategies that assume stationary return dynamics suffer significant drawdowns during regime shifts, as the 2008 Global Financial Crisis, the 2020 COVID-19 crash, and the 2022 interest rate hiking cycle demonstrated. Even diversified portfolios that hold all sectors with equal or risk-proportional weights cannot avoid the synchronous drawdowns that characterize systemic stress events.

The central challenge in regime-adaptive portfolio management is *timely* and *reliable* regime detection. Individual detectors operate on different time-scales and suffer from distinct failure

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modes: short-term indicators produce false alarms during idiosyncratic volatility, while long-term models react too slowly to protect against rapid drawdowns. The secondary challenge is translating regime signals into portfolio actions that actually reduce risk: a system that detects stress but merely redistributes weight among risky assets—rather than moving to cash—provides limited drawdown protection.

This paper addresses both challenges. We propose an ensemble architecture that fuses four complementary detectors through a Takagi-Sugeno fuzzy inference system (**?**), combined with an adaptive basket allocation framework where implicit cash allocation serves as the primary risk-reduction mechanism.

Our contributions are fourfold:

1. A multi-scale detector ensemble spanning ultra-short (5-day) to long-term (252+ day) horizons, with CUSUM operating on z-score-standardized returns for scale invariance and Markov-switching using a recursive Hamilton filter for real-time out-of-sample updates.

2. A fuzzy aggregation framework that learns optimal detector weights and membership functions by minimizing the Brier score against realized drawdowns, with bounded optimization and minimum weight floors to prevent detector suppression.

3. A tercile-based basket construction methodology that classifies sector ETFs using GARCH(1,1)-t conditional volatility and Ornstein-Uhlenbeck recovery dynamics, with graduated Core-basket response to extreme stress.

4. An implicit cash mechanism where stress-induced weight reduction produces sub-unity portfolio allocation, with the deficit earning the risk-free rate rather than being redistributed to other risky assets.

2 Related Work

? introduced Markov-switching models for business cycle analysis, establishing the foundation for regime-based asset allocation. **?** extended this framework to international asset allocation, demonstrating that regime-switching models can improve out-of-sample portfolio performance when properly calibrated.

? demonstrated the effectiveness of fuzzy logic systems for financial decision-making under uncertainty. The Takagi-Sugeno framework (**?**) provides a principled approach to combining multiple information sources with calibratable membership functions. Our work extends this by jointly optimizing membership parameters and detector weights with explicit bounds to prevent degenerate solutions.

? showed that regime-based asset allocation outperforms static strategies when the regime model is properly calibrated. **?** demonstrated that simple timing strategies based on inverse-volatility weighting outperform naïve diversification. Our framework builds on inverse-volatility weighting as the base allocation and adds regime-dependent scaling with an implicit cash buffer.

? introduced the CUSUM procedure for sequential change detection. **?** proposed the PELT algorithm for optimal change-point detection with linear computational cost. **?** derived average run length approximations for setting CUSUM decision thresholds. We apply CUSUM to z-score-standardized returns to achieve scale-invariant detection thresholds.

3 System Architecture

Figure **??** presents the complete system architecture. The framework operates in four layers: data acquisition, multi-scale detection with fuzzy aggregation, asset characterization and classification, and adaptive portfolio construction with implicit cash allocation. All parameters are re-estimated at each quarterly walk-forward step using only past data, ensuring zero lookahead bias.

architecture.pdf

Figure 1: System architecture. Four detectors spanning 5-day to 252+-day horizons feed into a Brier-score-optimized fuzzy aggregator. The composite stress probability drives adaptive basket allocation with implicit cash as the risk-reduction mechanism. All parameters are re-estimated at each walk-forward step.

4 Methodology

4.1 Multi-Scale Detector Ensemble

We deploy four detectors targeting complementary time-scales (Figure **??** illustrates their coordinated response during the COVID-19 crash). All parameters are estimated from training data at

each walk-forward step.

4.1.1 CUSUM Sequential Detector (5–10 day)

Following **?**, we maintain upward and downward CUSUM accumulators on *z-score-standardized* returns:

$$z_t = \frac{r_t - \hat{\mu}_0}{\hat{\sigma}_{\text{train}}} \tag{1}$$

$$S_t^+ = \max(0,\ S_{t-1}^+ + z_t - k) \tag{2}$$

$$S_t^- = \max(0,\ S_{t-1}^- - z_t - k) \tag{3}$$

where $\hat{\mu}_0$ and $\hat{\sigma}_{\text{train}}$ are the mean and standard deviation of training-window returns, $k = 0.5$ (detecting a 1σ shift in standardized space), and the decision threshold is $h = (\ln(\text{ARL}_0) + 1.166)/\delta$ with $\delta = 1$ per **?**, yielding $h \approx 6.70$ for a target average run length of 252 trading days. The signal is $\min(\max(S_t^+, S_t^-)/h,\ 1)$.

Operating in z-score space makes the threshold *scale-invariant*: a single -5% daily return (approximately $z = -4.2$) produces a signal of ~ 0.55, while three consecutive -2% days ($z \approx -1.7$ each) yield a signal of ~ 0.54. Under the original raw-return formulation, the threshold was approximately 558, requiring over 12,000 consecutive crash days to trigger—rendering the detector inoperative.

4.1.2 EWMA Volatility Crossover (21–42 day)

Two EWMA volatility estimators with decay factors λ_{fast} and λ_{slow} are calibrated via maximum likelihood estimation (**?**):

$$\hat{\sigma}_{\cdot,t}^2 = \lambda_\cdot\, \hat{\sigma}_{\cdot,t-1}^2 + (1 - \lambda_\cdot)\, r_t^2 \tag{4}$$

with $\lambda_{\text{fast}} \in [0.85, 0.96]$ and $\lambda_{\text{slow}} \in [0.95, 0.995]$. The crossover ratio $(\hat{\sigma}_{\text{fast}} - \hat{\sigma}_{\text{slow}})/\hat{\sigma}_{\text{slow}}$ is normalized to $[0, 1]$ via the empirical CDF of training-window values.

4.1.3 Hamilton Markov-Switching with Recursive Filter (63–126 day)

We fit a 2-state Markov-switching regression (**?**) on weekly SPY returns:

$$r_t \mid S_t = i \ \sim\ \mathcal{N}(\mu_i, \sigma_i^2), \quad i \in \{0, 1\} \tag{5}$$

During the out-of-sample window, rather than holding the filtered probability constant, we implement a *recursive Hamilton filter update*. Daily returns are accumulated into a pseudo-weekly buffer, and every 5 trading days the filter is updated via the standard predict–likelihood–update cycle:

$$\boldsymbol{\xi}_{t|t} = \frac{\boldsymbol{\xi}_{t|t-1} \odot \boldsymbol{\eta}}{\mathbf{1}^\top (\boldsymbol{\xi}_{t|t-1} \odot \boldsymbol{\eta})} \tag{6}$$

where $\boldsymbol{\xi}_{t|t-1} = \mathbf{P}^\top \boldsymbol{\xi}_{t-1|t-1}$ is the prediction step using the transition matrix $\mathbf{P}$, and $\eta_j = \phi(r_t^{(w)}; \mu_j, \sigma_j^2)$ are state-conditional likelihoods of the pseudo-weekly return. The signal is $P(S_t = \text{stress} \mid r_1, \ldots, r_t)$.

4.1.4 Structural Break Detection (252+ day)

Using the PELT algorithm (**?**), we detect structural breaks in the cumulative return series with a BIC-calibrated penalty $\lambda = \ln(n) \cdot \hat{\sigma}^2$. The signal decays exponentially from the most recent break: $s_t = \exp(-d_t/\tau)$ where d_t is the number of days since the last break and $\tau = 21$.

4.2 Fuzzy Aggregation Framework

Each detector signal $x_i \in [0, 1]$ passes through a calibrated sigmoid membership function (**?**):

$$\mu_{\text{stress}}(x_i) = \frac{1}{1 + \exp(-a_i(x_i - c_i))} \tag{7}$$

The composite stress probability is:

$$P(\text{stress}) = \sum_{i=1}^{4} w_i \cdot \mu_{\text{stress}}(x_i), \quad \sum w_i = 1,\ w_i \geq w_{\min} \tag{8}$$

Parameters $\{a_i, c_i, w_i\}$ are jointly optimized by minimizing the Brier score (**?**):

$$\mathcal{L}_{\text{Brier}} = \frac{1}{T} \sum_{t=1}^{T} \big(P_{\text{composite},t} - D_t\big)^2 \tag{9}$$

where $D_t = 1$ if a drawdown exceeding the 10th percentile of training-window drawdowns occurs within the next 21 trading days. Optimization uses L-BFGS-B with bounds: $a_i \in [1, 50]$, $c_i \in [0.05, 0.95]$, raw weights $\tilde{w}_i \in [-5, 5]$ (mapped through softmax). A minimum weight floor of $w_{\min} = 0.05$ is applied:

$$w_i = w_{\min} + (1 - 4w_{\min}) \cdot \text{softmax}(\tilde{w})_i \tag{10}$$

Figure **??** shows the calibrated membership functions and optimized detector weights.

4.3 Asset Characterization and Basket Construction

Each sector ETF is characterized by GARCH(1,1)-t conditional volatility (**??**) and Ornstein-Uhlenbeck recovery half-life (**?**). Figure **??** shows the resulting classification landscape. Sectors are assigned to three baskets using *tercile* boundaries on conditional volatility (33rd and 67th percentiles) combined with median half-life:

- **Basket A (Tactical):** Top or middle tercile vol with fast recovery. Binary liquidation above the calibrated entry threshold; re-entry below the exit threshold.

- **Basket B (Avoid):** Top tercile vol with slow recovery. Continuous de-risking: $w_i = w_i^{\text{base}} \cdot (1 - P(\text{stress}))$.
- **Basket C (Core):** Bottom tercile vol. Graduated linear scaling above $P(\text{stress}) = 0.7$:

$$w_i = w_i^{\text{base}} \cdot \max\left(0,\ \frac{1 - P(\text{stress})}{0.3}\right) \tag{11}$$

4.4 Implicit Cash Allocation

Base weights are computed using inverse-volatility weighting (**?**), *normalized to sum to 1.0 before any stress scaling is applied.* After basket-specific stress scaling:

$$W_{\text{total}} = \sum_i w_i \leq 1.0, \qquad w_{\text{cash}} = 1 - W_{\text{total}} \tag{12}$$

The deficit earns the daily risk-free rate. This design avoids a critical pitfall: when stressed assets are de-weighted and renormalized to sum to 1.0, the freed capacity is redistributed to remaining assets, potentially *increasing* concentration risk during stress. By allowing sub-unity exposure, the system achieves genuine risk reduction.

4.5 Walk-Forward Validation

The system is validated using expanding-window walk-forward analysis with a 504-day (2-year) minimum training window and 63-day (quarterly) out-of-sample steps. At each step, *all* parameters are re-estimated from training data only. Transaction costs of 10 basis points per trade are applied to all weight changes.

5 Results

5.1 Data

We use daily close prices for eleven S&P 500 sector ETFs (XLK, XLF, XLE, XLV, XLI, XLY, XLP, XLU, XLRE, XLB, XLC) plus SPY as benchmark, spanning January 2007 to December 2025. Fama-French five-factor data (**?**) provides the daily risk-free rate. The out-of-sample evaluation period covers 4,274 trading days from January 2009 to December 2025.

5.2 Portfolio Performance

Table **??** summarizes the out-of-sample results. The regime-adaptive strategy captures 98.5% of SPY's annualized return (11.61% vs. 11.79%) while reducing volatility by 26% (13.22% vs. 17.98%) and maximum drawdown by 53% (−19.60% vs. −41.71%).

Table 1: Walk-forward out-of-sample performance (2009–2025)

Metric	**Regime-Adaptive**	**SPY B&H**	**Equal-Weight**
Ann. Return	11.61%	11.79%	8.52%
Ann. Volatility	13.22%	17.98%	16.05%
Sharpe Ratio	**0.78**	0.58	0.45
Sortino Ratio	**0.75**	0.54	0.42
Calmar Ratio	**0.59**	0.28	0.18
Max Drawdown	**−19.60%**	−41.71%	−46.40%
Max DD Duration	**226d**	512d	530d
Ann. Turnover	4.01	—	—
Regime Transitions	116	—	—
Hit Rate	91.6%	—	—
False Alarm Rate	8.4%	—	—

The Sharpe ratio of 0.78 exceeds SPY's 0.58 by 34%, driven entirely by volatility reduction rather than return enhancement. The Calmar ratio more than doubles that of the benchmark (0.59 vs. 0.28), indicating substantially improved drawdown-adjusted performance. Figure **??** shows the cumulative equity curves alongside the drawdown profiles, and Figure **??** presents the risk-adjusted metric comparisons.

equity_curves.pdf

Figure 2: Out-of-sample equity curves (top) and drawdown profiles (bottom), 2009–2025. The regime-adaptive strategy matches SPY's terminal wealth while exhibiting substantially shallower drawdowns. The COVID-19 drawdown reached -19.6% versus -41.7% for SPY.

performance_bars.pdf

(a) Risk-adjusted performance ratios

max_drawdown.pdf

(b) Maximum drawdown comparison

Figure 3: Performance comparison across all three strategies. The regime-adaptive approach achieves superior risk-adjusted returns (a) with a maximum drawdown 53% smaller than SPY (b).

5.3 Rolling Performance Stability

Figure **??** shows the 252-day rolling Sharpe ratio for both the regime-adaptive strategy and SPY. The strategy maintains a consistently higher rolling Sharpe, with particularly large outperformance during and immediately after stress periods (2009–2010, 2020, 2022). The rolling Sharpe rarely drops below zero for the adaptive strategy, whereas SPY experiences extended negative-Sharpe periods during drawdowns.

rolling_sharpe.pdf

Figure 4: 252-day rolling Sharpe ratio. The regime-adaptive strategy maintains consistently higher risk-adjusted returns, with the largest outperformance occurring during and after stress periods.

5.4 Annual Return Decomposition

Figure **??** shows the year-by-year return comparison. The strategy captures most of SPY's upside in bull years while significantly outperforming in drawdown years (2020, 2022). This asymmetric profile—participating in gains, avoiding losses—is the hallmark of an effective regime-responsive system.

annual_returns.pdf

Figure 5: Annual return comparison. The strategy participates in bull markets while significantly outperforming during drawdown years (2020, 2022), demonstrating the value of regime-responsive cash allocation.

5.5 Multi-Scale Detector Response

Figure **??** illustrates the coordinated detector response during the COVID-19 crash. The CUSUM detector fires first (signal ≈ 0.55 on the initial -3.4% day on February 24), providing rapid detection within a single trading day. The EWMA crossover follows as the volatility regime shifts over the subsequent week. The recursive Markov filter reaches 0.98 stress probability within 5 trading days. The structural break detector provides confirmatory long-horizon evidence over the following weeks.

detector_covid.pdf

Figure 6: Multi-scale detector response during the COVID-19 crash (January–June 2020). Top panel: SPY price. Lower panels: individual detector signals. The CUSUM fires within one day, followed by EWMA (days), Markov (weeks), and structural break (months)—each contributing at its native time-scale.

5.6 Detector Contribution Analysis

The Brier-score-optimized weights with minimum floor reveal the relative importance of each time-scale (Figure **??**). CUSUM receives the dominant weight ($\sim$78%), reflecting the value of rapid detection in the z-score formulation. The structural break detector contributes $\sim$12%, capturing long-horizon regime changes. EWMA and Markov-switching each contribute at or near the 5%

floor.

membership_weights.pdf

Figure 7: Left: calibrated sigmoid membership functions showing each detector's activation profile. Right: Brier-score-optimized detector weights with 5% minimum floor (dashed line). CUSUM dominates at ∼78%, consistent with the value of rapid z-score-based detection.

5.7 Cash Allocation Dynamics

The implicit cash mechanism is the primary driver of drawdown reduction. Figure **??** shows the composite stress probability alongside the portfolio's cash allocation over the full backtest period. Stress spikes trigger proportional cash increases, with the most dramatic response during COVID-19 and the 2022 rate-hiking drawdown.

stress_cash.pdf

Figure 8: Composite stress probability (top) and portfolio cash allocation (bottom) over the full out-of-sample period. Stress spikes produce proportional increases in cash, with the largest allocations during COVID-19 (2020) and the rate-hiking cycle (2022). The dashed line marks the 0.7 threshold above which Core basket scaling activates.

Figure **??** provides a detailed view of the COVID-19 period. The portfolio's equity exposure dropped to 21.1% (78.9% implicit cash) within approximately one week of the drawdown onset. Recovery re-entry occurred gradually through April–May 2020 as $P(\text{stress})$ declined, consistent with the Ornstein-Uhlenbeck half-life estimates for the fastest-recovering sectors.

covid_zoom.pdf

Figure 9: COVID-19 case study (February–July 2020). Top: relative performance of the strategy vs. SPY. Bottom: portfolio equity and cash weights. The strategy moved to ∼79% cash within days of the crash onset and gradually re-entered during recovery.

5.8 Basket Classification

Figure **??** shows the sector ETF classification landscape at a representative rebalance date. The tercile-based classification produces a typical distribution of 3–4 assets in Basket A (Tactical), 1–3 in Basket B (Avoid), and 4–6 in Basket C (Core), providing 5–7 actively managed positions at each rebalance. Energy (XLE) and Financials (XLF) frequently appear in the higher-volatility baskets, consistent with their conditional risk profiles.

basket_scatter.pdf

Figure 10: Sector ETF basket classification based on GARCH conditional volatility (y-axis) and OU recovery half-life (x-axis). Tercile boundaries on volatility and median half-life determine basket assignment. Higher-volatility sectors with fast recovery (Tactical) are actively traded; those with slow recovery (Avoid) are continuously de-risked.

6 Discussion

6.1 The Normalization Problem

A critical design insight is that portfolio weight normalization can negate regime-responsive behavior. Inverse-volatility weights for N assets produce raw sums of $\sim$20–30. When stressed assets are de-weighted and the result is normalized to sum to 1.0, the freed capacity is redistributed proportionally to surviving positions—including Core assets. This means the portfolio *increases* its allocation to these assets during stress, the opposite of risk reduction. Our approach normalizes base weights to 1.0 *before* stress scaling, so that post-scaling sums below 1.0 represent genuine cash allocation.

6.2 Detector Scale Invariance

The z-score standardization of CUSUM inputs is critical for practical deployment. With raw returns, the Siegmund threshold scales inversely with the standard deviation ($h \propto 1/\sigma$), producing thresholds of $\sim$558 for typical equity returns ($\sigma \approx 0.012$). Daily accumulator increments of $\sim$0.04 require thousands of consecutive crash days to trigger. In z-score space, the threshold is $\sim$6.70 regardless of return scale, and a single -5% day produces a meaningful signal of 0.55.

6.3 Limitations

The framework has several limitations. First, the annualized turnover of $4.01\times$ creates transaction cost drag, though the 10 bps assumption is conservative for liquid ETFs. Second, the quarterly recalibration frequency may lag rapidly evolving dynamics; event-driven recalibration could improve responsiveness. Third, the CUSUM detector's $\sim$78% weight dominance suggests the remaining detectors contribute primarily through ensemble diversity. Fourth, the backtest spans a single macroeconomic epoch (post-GFC monetary expansion); performance in structurally different interest rate environments requires further validation.

6.4 Robustness

All parameters are re-estimated at each walk-forward step using only past data. The bounded L-BFGS-B optimization with minimum weight floors prevents convergence to degenerate single-detector solutions. The implicit cash mechanism provides a natural risk-reduction pathway that does not depend on predicting the *type* of stress, only its *presence*.

7 Conclusion

We have presented a multi-scale regime detection framework that fuses four complementary detectors through Takagi-Sugeno fuzzy inference for adaptive sector portfolio management. The walk-forward validated results demonstrate a Sharpe ratio of 0.78 with a maximum drawdown of -19.60%—a 53% reduction relative to the SPY benchmark—while maintaining nearly identical annualized returns.

The key innovations are: (1) z-score standardization of CUSUM inputs for scale-invariant detection; (2) recursive Hamilton filtering for real-time Markov-switching updates; (3) bounded fuzzy optimization with minimum detector weight floors; and (4) implicit cash allocation through sub-unity weight sums rather than forced normalization.

The framework's success lies not in predicting market direction but in reliably detecting the *onset* of stress and responding with genuine risk reduction. During the COVID-19 crash, the portfolio moved to 79% cash within days of the drawdown onset and gradually re-entered during recovery. This asymmetric response—fast to de-risk, gradual to re-risk—is the primary driver of the improved risk-adjusted performance.

Future work may explore: (1) event-driven recalibration triggered by detector alerts; (2) cross-sectional regime signals from equity factor models; (3) extension to multi-asset-class allocation; and (4) incorporation of options-based tail hedging as a complement to the implicit cash mechanism.

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